

Appendix 1

Two formulae of Clebsch-Gordan are:

$$\begin{aligned} r_x^m s_y^n &= \sum_k \frac{\binom{m}{k} \binom{n}{k}}{\binom{m+n-k+1}{k}} (\overline{rs})_{y^{n-k}}^k (xy)^k, \\ (\overline{rs})_{y^{n-k}}^k &= \sum_k (-1)^k \frac{\binom{m}{k} \binom{n}{k}}{\binom{m+n}{k}} (rs)^\nu r_x^{m-\nu} s_y^{n-\nu} (xy)^\nu. \end{aligned} \tag{1}$$

I prefer to formulate it in terms of matrices. Assume $m \leq n$. Let $A = (a_{ij})$, with $0 \leq i, j \leq m$, where

$$a_{ij} = 0 \text{ if } i > j, 1 \leq i \leq j, \frac{\binom{m-i}{j-i} \binom{n-i}{j-i}}{\binom{m+n-i-j+1}{j-i}} i \text{ if } i < j. \text{ Similarly, let } B = (b_{ij}), \text{ with the same range of indices, where}$$

$$b_{ij} = 0 \text{ if } i > j, 1 \leq i \leq j, (-1)^{i+j} \frac{\binom{m-i}{j-i} \binom{n-i}{j-i}}{\binom{m+n-2i}{j-i}} i \text{ if } i < j.$$

Then Gordan's inversion formula amounts to the non-obvious assertion that B is the inverse of A .

I want to record one observation I have made about Gordan's coefficients. The matrix A is an upper triangular unipotent matrix. Let $C = (c_{ij})$ be the logarithm of A . Then for $j > i$ we have

$$C_{ij} = \kappa_{j-i} A_{ij} \prod_{k=0}^{j-i-2} \frac{1}{n+m-2i-k},$$

where $\kappa = \{\kappa_r\}_{r=1,2,3,\dots}$ is a sequence of integers independent of m, n . We present the first 20 terms of the sequence κ below.

r	κ_r
1	1
2	1
3	3
4	14
5	80
6	468
7	2268
8	10224
9	313632
10	9849600
11	21954240
12	-8894136960
13	-105857556480
14	20609598562560
15	650835095904000
16	-80028503341516800

17 -5018759207362252800
18 503681435808239001600
19 56090762228110443724800
20 -4811020757268709294080000